Practical Exercise 2: Circuit Design

Part A Calculations

(a) Series Circuit

The resistors are in series:

$$R_T = R_1 + R_2 + R_3$$

1. Total Resistance

$$R_T = 100\Omega$$

2. Total Current

Using Ohm's Law:

$$I = \frac{V}{R}$$

$$I = \frac{125V}{100\Omega} = 1.25A$$

Total Current

$$I_T = 1.25A$$

3. Current Through Each Resistor

In a series circuit, current is the same everywhere:

$$I_1 = I_2 = I_3 = 1.25A$$

4. Voltage Drop Across Each Resistor

Using:

$$V = IR$$

Across 20Ω

$$V_1 = (1.25A)(20\Omega) = 25V$$

Across 30Ω

$$V_2 = (1.25A)(30\Omega) = 37.5V$$

Across 50Ω

$$V_3 = (1.25A)(50\Omega) = 62.5V$$

5. Power Dissipated

Using Watt's Law:

$$P = VI$$

20Ω

$$P_1 = 25V \times 1.25A = 31.25W$$

30Ω

$$P_2 = 37.5V \times 1.25A = 46.875W$$

50Ω

$$P_3 = 62.5V \times 1.25A = 78.125W$$

(b) Parallel Circuit

Resistors:

- 20Ω
- 100Ω
- 50Ω

1. Total Resistance

For parallel circuits:

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = 0.08$$

$$R_T = \frac{1}{0.08}$$

$$R_T = 12.5\Omega$$

2. Total Current

$$I_T = \frac{V}{R_T}$$

$$I_T = \frac{125V}{12.5\Omega} = 10A$$

3. Current Through Each Resistor

In parallel, voltage is the same across each branch.

Through 20Ω

$$I_1 = \frac{125}{20} = 6.25A$$

Through 100Ω

$$I_2 = \frac{125}{100} = 1.25A$$

Through 50Ω

$$I_3 = \frac{125}{50} = 2.5A$$

4. Voltage Drop

In parallel:

$$V_1 = V_2 = V_3 = 125V$$

5. Power Dissipated

20Ω

$$P_1 = 125 \times 6.25 = 781.25W$$

100Ω

$$P_2 = 125 \times 1.25 = 156.25W$$

50Ω

$$P_3 = 125 \times 2.5 = 312.5W$$